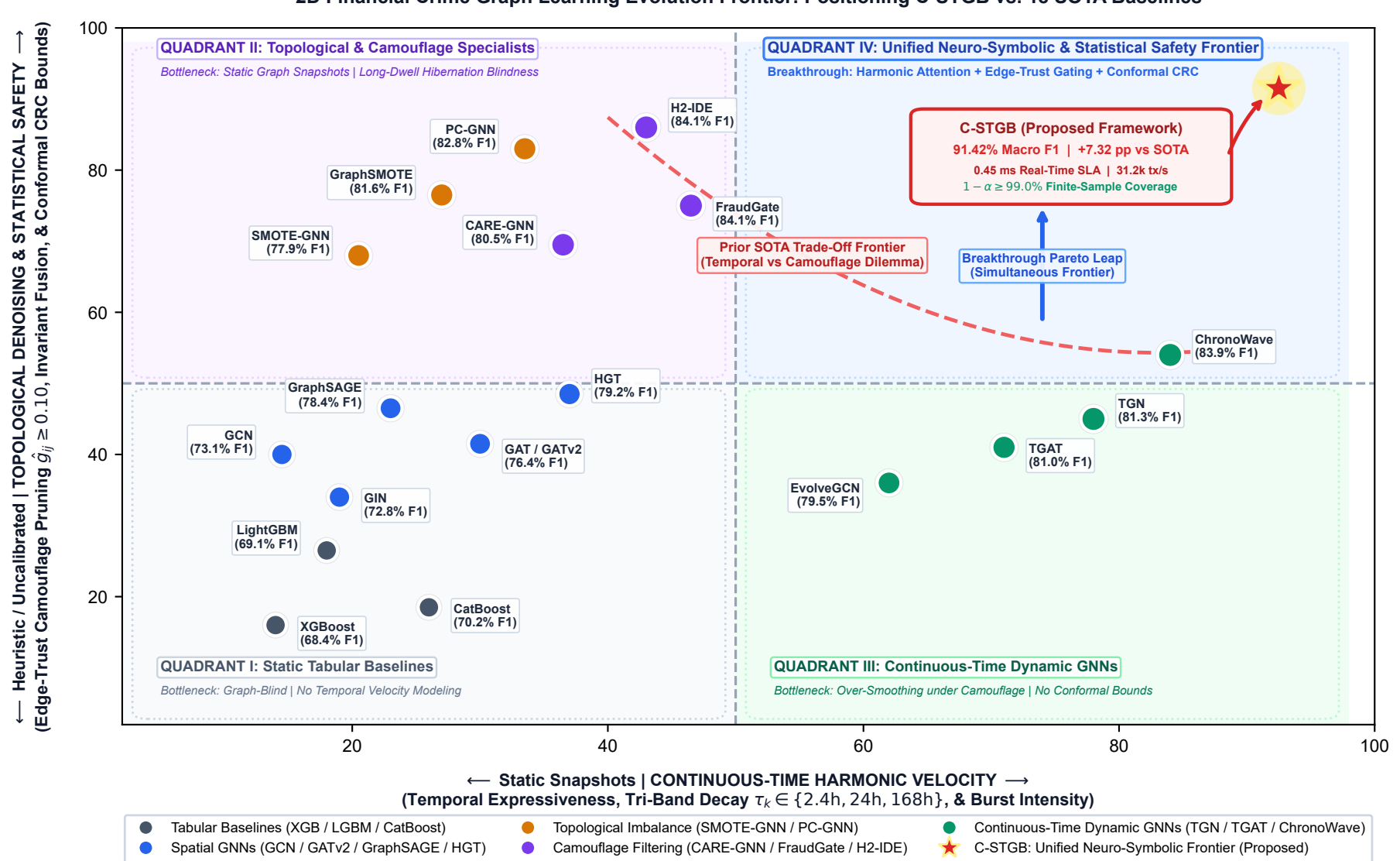

2D Financial Crime Graph Learning Evolution Frontier: Positioning C-STGB vs. 18 SOTA Baselines
← Heuristic / Uncalibrated | TOPOLOGICAL DENOISING & STATISTICAL SAFETY →
(Edge-Trust Camouflage Pruning $\hat{g}_{ij} \geq 0.10$, Invariant Fusion, & Conformal CRC Bounds)
QUADRANT II: Topological & Camouflage Specialists
Bottleneck: Static Graph Snapshots | Long-Dwell Hibernation Blindness
QUADRANT IV: Unified Neuro-Symbolic & Statistical Safety Frontier
Breakthrough: Harmonic Attention + Edge-Trust Gating + Conformal CRC
C-STGB (Proposed Framework)
91.42% Macro F1 | +7.32 pp vs SOTA
0.45 ms Real-Time SLA | 31.2k tx/s
$1 - \alpha \geq 99.0\%$ Finite-Sample Coverage
H2-IDE (84.1% F1)
PC-GNN (82.8% F1)
GraphSMOTE (81.6% F1)
SMOTE-GNN (77.9% F1)
CARE-GNN (80.5% F1)
FraudGate (84.1% F1)
Prior SOTA Trade-Off Frontier (Temporal vs Camouflage Dilemma)
Breakthrough Pareto Leap (Simultaneous Frontier)
ChronoWave (83.9% F1)
HGT (79.2% F1)
GraphSAGE (78.4% F1)
GCN (73.1% F1)
GAT / GATv2 (76.4% F1)
GIN (72.8% F1)
LightGBM (69.1% F1)
CatBoost (70.2% F1)
XGBoost (68.4% F1)
TGN (81.3% F1)
TGAT (81.0% F1)
EvolveGCN (79.5% F1)
QUADRANT I: Static Tabular Baselines
Bottleneck: Graph-Blind | No Temporal Velocity Modeling
QUADRANT III: Continuous-Time Dynamic GNNs
Bottleneck: Over-Smoothing under Camouflage | No Conformal Bounds
100
80
60
40
20
20
40
60
80
100
← Static Snapshots | CONTINUOUS-TIME HARMONIC VELOCITY →
(Temporal Expressiveness, Tri-Band Decay $\tau_k \in \{2.4\text{h}, 24\text{h}, 168\text{h}\}$, & Burst Intensity)
Tabular Baselines (XGB / LGBM / CatBoost)
Spatial GNNs (GCN / GATv2 / GraphSAGE / HGT)
Topological Imbalance (SMOTE-GNN / PC-GNN)
Camouflage Filtering (CARE-GNN / FraudGate / H2-IDE)
Continuous-Time Dynamic GNNs (TGN / TGAT / ChronoWave)
C-STGB: Unified Neuro-Symbolic Frontier (Proposed)